



**Northern Beaches
Secondary College**

Manly Selective Campus

2014 HSC –Trial Examination

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes.
- Working time – 3 hours .
- Write using blue or black pen.
- Board-approved calculators and templates may be used.
- All necessary working should be shown in every question.
- Multiple choice questions to be completed on the special answer page.
- Each free response questions to be completed in separate booklets.
- If using more than one booklet per question, number booklet “1 of ___”

Total marks – 100 marks

- Attempt Questions 1-16
- Multiple Choice – answer question on answer sheet provided.
- Multiple Choice – 1 mark per question
- Short Answer questions – marks as indicated.

MULTIPLE CHOICE SECTION.

Answer the following questions on the answer sheet provided.

Q1. If $z = 1 + 2i$ and $\omega = 3 - i$ then $z - \overline{\omega}$ is:

$i - 2$

$4 + i$

$3i - 2$

$4 + 3i$

Q2. The directrices of the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ are :

A) $x = \pm \frac{16\sqrt{7}}{7}$

B) $x = \pm \frac{\sqrt{7}}{16}$

C) $x = \pm \frac{16}{5}$

D) $x = \pm \frac{5}{16}$

Q3. Let α, β, γ be the roots of the equation $x^3 + x - 1 = 0$. The polynomial equation with roots $2\alpha\beta, 2\alpha\gamma, 2\beta\gamma$ is:

A $x^3 + 2x^2 + 2 = 0$

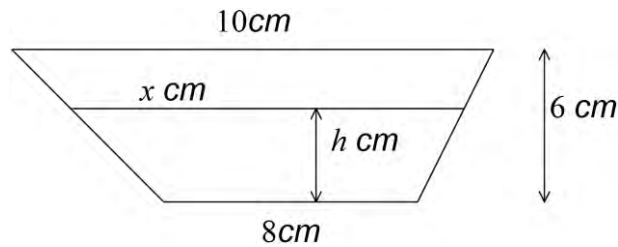
B $x^3 - 2x^2 - 2 = 0$

C $x^3 - 2x^2 - 8 = 0$

D $x^3 - 2x^2 + 8 = 0$

- Q4. If $f(x) = \frac{x(x-1)}{x-2}$, which of the following lines will be an asymptote to the graph $y = f(x)$?
- A) $y = x + 1$
 - B) $y = x - 2$
 - C) $y = x - 1$
 - D) $y = 0$
- Q5. $P(z)$ is a polynomial of degree 4. Which one of the following statements must be false?
- A $P(z)$ has no real roots.
 - B $P(z)$ has 1 real root and 3 non-real roots
 - C $P(z)$ has 2 real roots and 2 non-real roots
 - D $P(z)$ has 4 non-real roots
- Q6. For a certain function $y = f(x)$, the function $y = f(|x|)$ is represented by :
- A) A reflection of $y = f(x)$ in the y -axis
 - B) A reflection of $y = f(x)$ in the x -axis.
 - C) A reflection of $y = f(x)$ in the x axis for $y \geq 0$
 - D) A reflection of $y = f(x)$ in the y axis for $x \geq 0$
- Q7. If ω is an imaginary cube root of unity, then $(1 + \omega - \omega^2)^7$ is equal to :
- A) 128ω
 - B) -128ω
 - C) $128\omega^2$
 - D) $-128\omega^2$

- Q8. The diagram shows a trapezium with an interval of x units drawn parallel to and h units above the base.



An expression for x in terms of h is given by:

- A) $x = 5 - \frac{h}{6}$
 B) $x = 8 + \frac{h}{6}$
 C) $x = 12 + \frac{h}{8}$
 D) $x = 8 + \frac{h}{3}$
- Q9. Given the eccentricity of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is e , then the eccentricity of the

ellipse $\frac{x^2}{a^2 + b^2} + \frac{y^2}{b^2} = 1$ is :

- A) $\sqrt{e}$
 B) $\frac{1}{e}$
 C) e
 D) e^2
- Q10. The roots of the polynomial $4x^3 + 4x - 5 = 0$ are α, β, γ . The value of $(\alpha + \beta - 3\gamma)(\beta + \gamma - 3\alpha)(\alpha + \gamma - 3\beta)$ is

- A) -80
 B) -16
 C) 16
 D) 80

FREE RESPONSE SECTION – answer each question in a separate booklet

Question 11 START A NEW BOOKLET

15 marks

- a) (i) Find the Cartesian equation of the locus of the point z in the complex plane given that $\operatorname{Re}(z) = |z - 2|$ (2)
- (ii) Sketch the locus of z . (1)
- b) Let $P(z)$ be a point in the complex plane such that $|z - i| = \frac{1}{2}$. Find the maximum value of $\operatorname{Arg}(z)$ (2)
- c) (i) Express $\sqrt{8 + 6i}$ in the form $x + iy$ where $x > 0$ (2)
- (ii) Solve the equation $z^2 + 2(1 + 2i)z - (11 + 2i) = 0$ (2)
- d) Let $f(x)$ be a continuous function for $-5 \leq x \leq 10$ and let $g(x) = f(x) + 2$.
If $\int_{-5}^{10} f(x) dx = 4$, show that $\int_{-5}^{10} g(u) du = 34$ (3)
- e) Use integration by parts to find $\int_0^1 \frac{\cos^{-1} x}{\sqrt{1+x}} dx$ (3)

Question 12 START A NEW BOOKLET

15 marks

- a) (i) Prove that any equation of the form

$$x^3 - mx^2 + n = 0$$

where $m, n \neq 0$ cannot have a triple zero. (2)

- (ii) If the equation has a double zero, find the relation between m and n (2)

- b) Use the table of standard integrals to find $\int \frac{2x}{\sqrt{x^4 + 16}} dx$ (2)

- c) The equation $2x^2 - kx + 17 = 0$ has one zero, α , such that $\operatorname{Re}(\alpha) = \frac{5}{2}$.
If k is real, find the other zero of the equation and the value of k . (2)

- d) (i) Use the substitution $t = \tan \frac{x}{2}$ to prove

$$\int_0^{\frac{\pi}{2}} \frac{dx}{2 + 2\sin x} = \frac{1}{2} \quad (3)$$

- (ii) Use the substitution $u = 2a - x$ to prove

$$\int_0^{2a} f(x) dx = \int_0^a \{f(x) + f(2a - x)\} dx \quad (2)$$

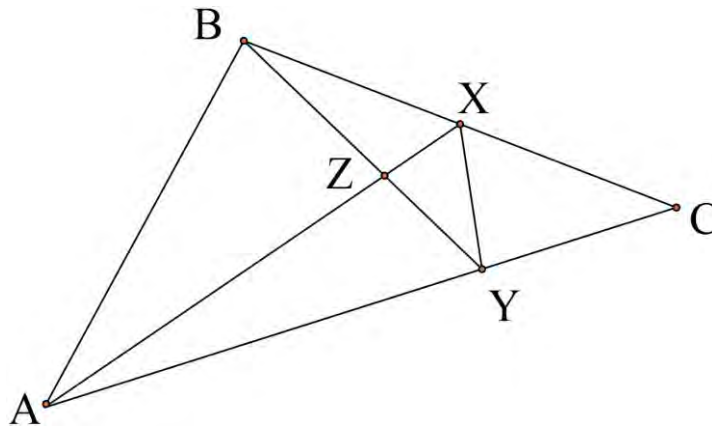
- (iii) Hence, or otherwise, evaluate

$$\int_0^{\pi} \frac{x}{2 + 2\sin x} dx \quad (2)$$

Question 13 START A NEW BOOKLET

15 marks

a)



X and Y are points on sides BC and AC of $\triangle ABC$.

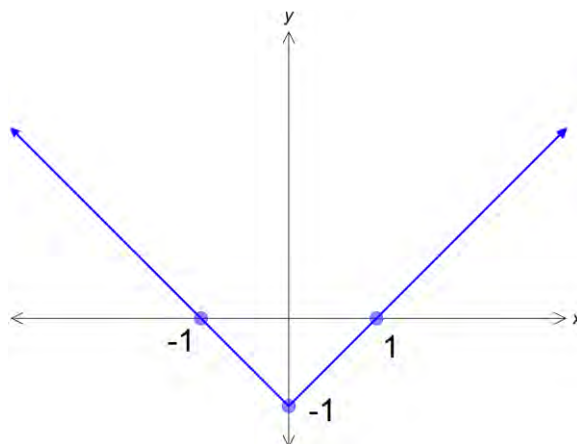
$\angle AXC = \angle BYC$ and $BX = XY$

- (i) Copy the diagram into your answer booklet, clearly showing the above information. (Your diagram must be at least $\frac{1}{4}$ page in size) (1)
- (ii) Prove $ABXY$ is a cyclic quadrilateral. (2)
- (iii) Hence, or otherwise, prove AX bisects $\angle BAC$. (3)

Question 13 continued on next page

Question 13 continueda

- b) Consider the function $f(x) = |x| - 1$ shown below.



Using diagrams, of at least $\frac{1}{4}$ page, sketch the following functions.
Show all essential features on your diagram.

- | | | |
|-------|--------------------|-----|
| (i) | $y = 1 - f(x)$ | (1) |
| (ii) | $y = x \cdot f(x)$ | (2) |
| (iii) | $ y = f(x)$ | (2) |
| (iv) | $y = e^{f(x)}$ | (2) |
| (v) | $y = \sqrt{f(x)}$ | (2) |

Question 14 START A NEW BOOKLET

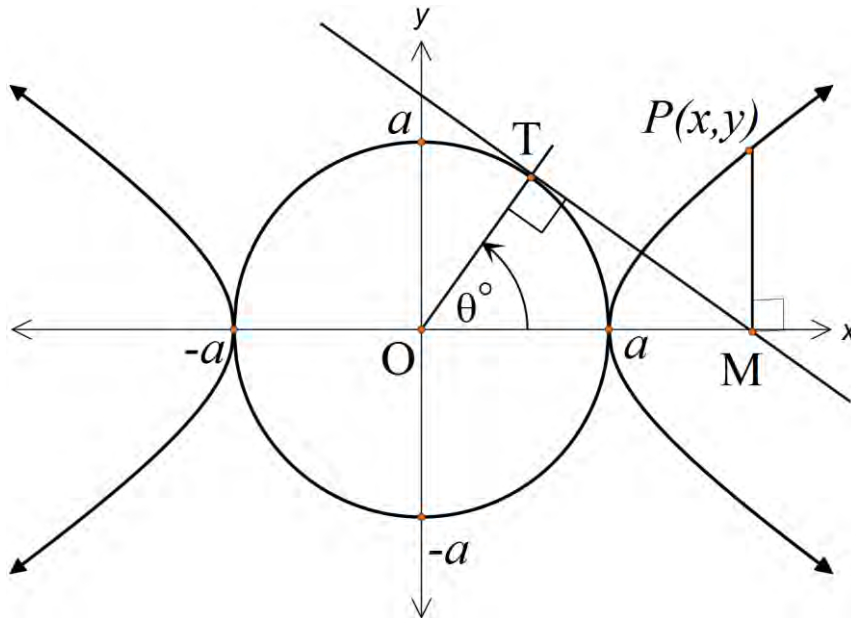
15 marks

- a) (i) Show that $(2 + i)$ is a root of $x^3 - 11x + 20 = 0$ (1)
- (ii) Hence, or otherwise solve $x^3 - 11x + 20 = 0$ (2)
- b) A, B, C and D are the vertices, in clockwise order, of a square.
- (i) Given that A and C represent the points $2 + 2i$ and $4 + 2i$ respectively, find the coordinates of B and D . (2)
- (ii) If the square $ABCD$ is rotated anticlockwise through 90° about the origin, find the coordinates of the new position of the point A . (1)
- c) Consider chords of the hyperbola $xy = c^2$ which pass through the point $A(6c, 4c)$
- (i) Find the equation of the chord passing through the points $P\left(cp, \frac{c}{p}\right)$ and $Q\left(cq, \frac{c}{q}\right)$. Express your answer in general form. (2)
- (ii) Show that $4pq + 6 = p + q$ (1)
- (iii) Find the equation of the locus of the midpoints of the chords PQ . (2)
- d) (i) Find $\frac{dy}{dx}$ for the implicit function $x^2 - y^2 + xy + 5 = 0$. (2)
- (ii) Find the x coordinate of the points of the curve $x^2 - y^2 + xy + 5 = 0$ where the tangents have a zero gradient. (2)

Question 15 START A NEW BOOKLET

15 marks

- a) The diagram below shows the hyperbola $H: \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and the circle $C: x^2 + y^2 = a^2$



The point $P(x, y)$ lies on the hyperbola H and the point T lies on the circle C and $\angle TOM = \theta$.

- (i) Show that P has the coordinates $(a \sec \theta, b \tan \theta)$. (2)
- (ii) The point Q lies on the hyperbola and has coordinates $(a \sec \phi, b \tan \phi)$.

Given that $\theta + \phi = \frac{\pi}{2}$ and $\theta \neq \frac{\pi}{4}$, show that the chord PQ has the equation

$$ay = b(\cos \theta + \sin \theta)x - ab \quad (4)$$

- (iii) Show that every chord PQ passes through a fixed point and find the coordinates of that point. (2)
- (iv) Show that, as θ approaches $\frac{\pi}{2}$, the chord PQ approaches a line parallel to an asymptote. (2)

Question 15 continued on next page

Question 15 continued

b) Let x_1, x_2 be positive real numbers.

(i) Show that $x_1 + x_2 \geq 2\sqrt{x_1 x_2}$ (1)

(ii) Hence, or otherwise, prove

$$\frac{x_1 + x_2 + x_3 + x_4}{4} > \sqrt[4]{x_1 x_2 x_3 x_4}$$

where x_1, x_2, x_3, x_4 are positive real numbers. (2)

(iii) By making the substitution $x_4 = \frac{x_1 + x_2 + x_3}{3}$ into the result in part (ii), show that

$$\frac{x_1 + x_2 + x_3}{3} > \sqrt[3]{x_1 x_2 x_3}$$

where x_1, x_2, x_3 are positive real numbers. (2)

Question 16 START A NEW BOOKLET

15 marks

- a) A solid has its base in the ellipse $E: \frac{x^2}{36} + \frac{y^2}{16} = 1$

If each section perpendicular to the major axis is an equilateral triangle, show that the volume of the solid is $128\sqrt{3}$ cubic units. (4)

- b) Let $I_n = \int_0^1 x(1-x^5)^n dx$ where $n \geq 0$ is an integer

(i) Show that $I_n = \frac{5n}{5n+2} I_{n-1}$ for $n \geq 1$ (3)

(ii) Show that $I_n = \frac{5^n n!}{2 \times 7 \times 12 \times \dots \times (5n+2)}$ for $n \geq 1$ (2)

(iii) Evaluate I_4 (1)

- c) Let $f(x), g(x)$ be continuously differentiable functions.

(i) Prove $\int \{f''(x)g(x) - f(x)g''(x)\} dx = f'(x)g(x) - f(x)g'(x)$ (2)

(ii) Hence, or otherwise, evaluate $\int_0^{\frac{\pi}{4}} \sin 3x \cos 2x dx$ (3)

(State your answer in simplest exact form)

End of Examination



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Answer sheet for multiple choice questions 1- 10

INSTRUCTIONS:

For Questions 1 to 10, place a cross in the box corresponding to your selected answer.

Student Number: _____

<i>Question #</i>	<i>(A)</i>	<i>(B)</i>	<i>(C)</i>	<i>(D)</i>
<i>1</i>				
<i>2</i>				
<i>3</i>				
<i>4</i>				
<i>5</i>				
<i>6</i>				
<i>7</i>				
<i>8</i>				
<i>9</i>				
<i>10</i>				

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

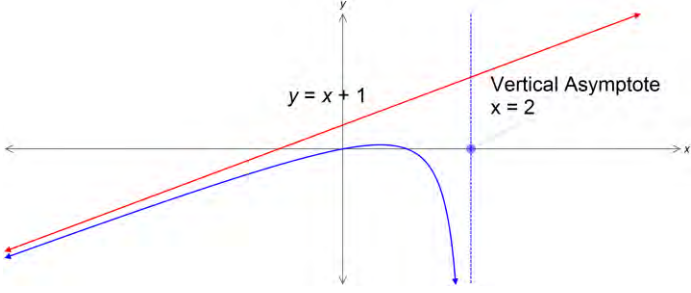
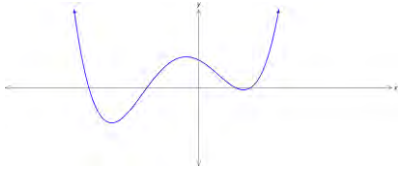
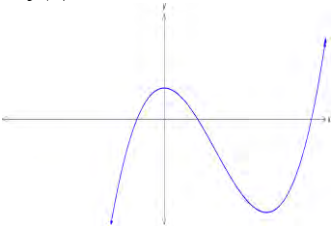
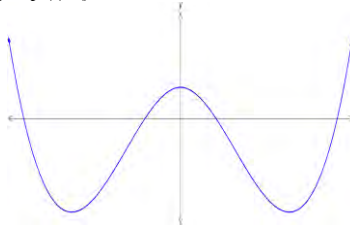
NOTE : $\ln x = \log_e x, \quad x > 0$

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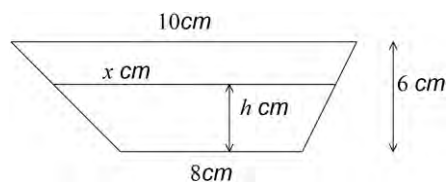
Question 1:

Q1	$z = 1 + 2i \quad \omega = 3 - i \quad \therefore \overline{\omega} = 3 + i$ $z - \overline{\omega} = 1 + 2i - (3 + i)$ $= i - 2$	A
Q2	$\frac{x^2}{16} + \frac{y^2}{9} = 1$ Directrices: $x = \pm \frac{a}{e} = \pm \frac{4}{e} \quad e = \sqrt{1 - \frac{b^2}{a^2}} = \frac{\sqrt{7}}{4}$ $x = \pm \frac{16}{\sqrt{7}} = \pm \frac{16\sqrt{7}}{7}$	A
Q3	$x^3 + x - 1 = 0$ $\therefore \alpha\beta\gamma = 1$ $2\alpha\beta = \frac{2\alpha\beta\gamma}{\gamma} = \frac{2}{\gamma}$ $2\alpha\gamma = \frac{2}{\beta}$ $2\beta\gamma = \frac{2}{\alpha}$ $\therefore x = \frac{2}{X}$ $\frac{8}{X^3} + \frac{2}{X} - 1 = 0$ $8 + 2X^2 - X^3 = 0$ $\therefore x^3 - 2x^2 - 8 = 0$	C

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Q4	$f(x) = \frac{x(x-1)}{x-2}$ 	A
Q5	It is not possible to have one real root and 3 unreal roots – see diagram. 	B
Q6	Resulting graph is reflection of positive x values in the y axis. $y=f(x)$  $y=f(x)$ 	D
Q7	$\begin{aligned}\omega^3 - 1 &= 0 \\ \therefore (\omega - 1)(\omega^2 + \omega + 1) &= 0 \quad \therefore (1 + \omega + \omega^2) = 0 \\ (1 + \omega - \omega^2)^7 &= (1 + \omega + \omega^2 - \omega^2 - \omega^2)^7 \\ &= (-2\omega^2)^7 \\ &= -128\omega^{14} = -128(\omega^{12}\omega^2) \\ &= -128\omega^2\end{aligned}$	D

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Short Method. – Linear method

x is a function of h ie.

$$x = ah + b$$

$$h = 0, x = 8$$

$$8 = 0 + b \Rightarrow b = 8$$

$$h = 6, x = 10$$

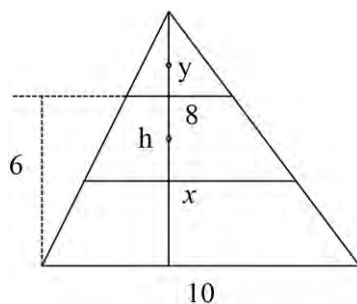
$$10 = 6a + 8$$

$$a = \frac{1}{3}$$

$$\therefore x = \frac{h}{3} + 8$$

Q8

Longer (tedious) method



$$\frac{y}{8} = \frac{y+6}{10} \Rightarrow y = 24$$

$$\frac{y}{8} = \frac{y+h}{x}$$

$$\frac{24}{8} = \frac{24+h}{x}$$

$$3x = 24 + h$$

$$x = 8 + \frac{h}{3}$$

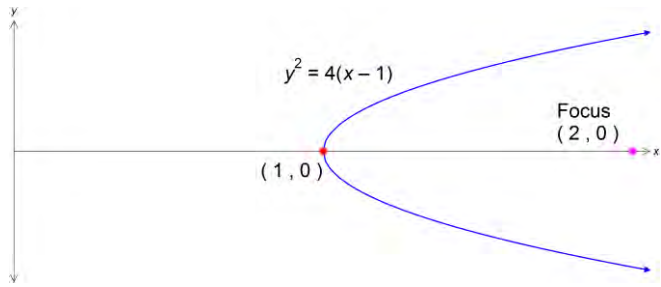
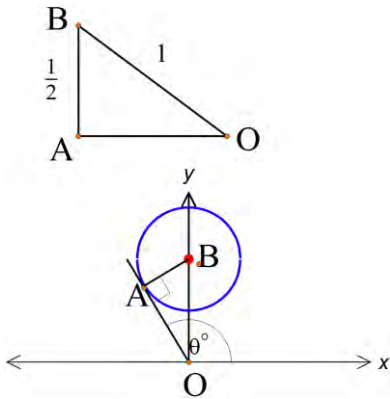
D

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Q9	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad e^2 = \left(\frac{b^2}{a^2} - 1 \right)$ $\therefore b^2 = a^2(e - 12)$ $\frac{x^2}{a^2 + b^2} + \frac{y^2}{b^2} = 1 \therefore E^2 = 1 - \frac{b^2}{a^2 + b^2}$ $\therefore E^2 = \frac{a^2 + b^2 - b^2}{a^2 + b^2} = \frac{a^2}{(a^2 + b^2)}$ $= \frac{a^2}{a^2 + a^2(e^2 - 1)} = \frac{1}{e^2}$ $\therefore E = \pm \frac{1}{e}$	B
Q10	$4x^3 + 4x - 5 = 0$ $\alpha + \beta + \gamma = 0 \quad \alpha\beta\gamma = \frac{5}{4}$ $(\alpha + \beta - 3\gamma)(\beta + \gamma - 3\alpha)(\alpha + \gamma - 3\beta)$ $= (\alpha + \beta + \gamma - 4\gamma)(\alpha + \beta + \gamma - 4\alpha)(\alpha + \beta + \gamma - 4\beta)$ $= -64\alpha\beta\gamma = -\frac{320}{4} = -80$	A

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Question 11

a-i	Let $z = x + iy \rightarrow \operatorname{Re}(z) = x$ $\operatorname{Re}(z) = z - 2$ $\therefore x = z - 2$ $x^2 = (x - 2)^2 + y^2$ $x^2 = x^2 - 4x + 4 + y^2$ $y^2 = 4(x - 1)$	2 marks – correct solution $x = z - 2$ 1 mark – recognising
a-ii		1 mark correct solution with (1,0)
b	$z - i = \frac{1}{2}$ $x^2 + (y - 1)^2 = \frac{1}{4}$  $\therefore \theta = \frac{\pi}{2} + \frac{\pi}{6} = \frac{2\pi}{3}$	2 marks – correct solution fully demonstrated. 1 mark – diagram showing circle and triangle correctly

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c-i	$\sqrt{8+6i} = x + iy$ $\therefore x^2 - y^2 = 8$ $2xy = 6 \Rightarrow y = \frac{3}{x}$ $x^2 + \frac{9}{x^2} = 8$ $x^4 + 9 = 8x^2$ $x^4 - 8x^2 + 9 = 0$ $(x^2 - 9)(x^2 + 1) = 0$ $\therefore x = \pm 3 \quad \text{nb } x \text{ is real and } x > 0$ $x = 3 \therefore y = 1$ $\sqrt{8+6i} = 3 + i$	2 marks correct solution 1 marks – correct except failure to recognise $x > 0$
c-ii	$0 = z^2 + 2(1+2i)z - (11+2i)$ $z = \frac{-2(1+2i) \pm \sqrt{4(1+2i)^2 + 4 \times (11+2i)}}{2}$ $= \frac{-2(1+2i) \pm 2\sqrt{1-4+4i+11+2i}}{2}$ $= (-1-2i) \pm \sqrt{8+6i}$ $= -1-2i \pm (3+i)$ $z = -1-2i+3+i = 2-i \text{ or } z = -1-2i-3-i = -4-3i$	2 marks – correct solution 1 mark – uses quadratic formula to arrive at non-simplified expression for z

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d	$\int_{-5}^{10} f(x) dx = 4$ $\therefore \int_{-5}^{10} g(x) dx = \int_{-5}^{10} f(x) + 2 dx = 4 + \left[2x \right]_{-5}^{10}$ $= 4 + 30 = 34$ if $u = x$ then $du = dx$ $nb \int g(x) dx = \int g(u) du = 34$	3 marks – correct solution – MUST include logic for $\int g(x) dx = \int g(u) du$ 2 marks – correct except for explanation. 1 mark – expansion of $g(x)$ integral
e	$\int_0^1 \frac{\cos^{-1} x}{\sqrt{1+x}} dx$ $u = \cos^{-1} x \quad v' = \frac{1}{\sqrt{1+x}}$ $u' = -\frac{1}{\sqrt{1-x^2}} \quad v = \frac{\sqrt{1+x}}{\frac{1}{2}}$ $I = \left[\cos^{-1} x \times 2(\sqrt{1+x}) \right]_0^1 + 2 \int_0^1 \frac{\sqrt{1+x}}{\sqrt{1+x}\sqrt{1-x}} dx$ $= \left(0 \times 2\sqrt{2} - 2 \times \frac{\pi}{2} \times 1 \right) + 2 \int_0^1 (1-x)^{-\frac{1}{2}} dx$ $= -\pi + 2 \times -2 \left[\sqrt{1-x} \right]_0^1$ $= -\pi - 4(0-1)$ $= -\pi + 4$	3 marks – correct solution 2 marks – correct value for initial term ie. $-\pi$ 1 mark – correct initial u, u', v and v'

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Question 12

a	$f(x) = x^3 - mx^2 + n$ $f'(x) = 3x^2 - 2mx$ $f''(x) = 6x - 2m = 0$ $\therefore x = \frac{m}{3}$ $\text{if } f'\left(\frac{m}{3}\right) = \frac{3m^2}{9} - \frac{2m^2}{3}$ $= -\frac{m^2}{3} \neq 0 \text{ if } m \neq 0$ therefore not possible to have triple root.	2 marks for correct solution 1 mark for $x = \frac{m}{3}$
b	$f(x) = x^3 - mx^2 + n$ $f'(x) = 3x^2 - 2mx$ $\therefore x = 0 \text{ or } x = \frac{2m}{3}$ $x = 0$ not a solution for $f(x)$ $f\left(\frac{2m}{3}\right) = \frac{8m^3}{27} - \frac{4m^3}{9} + n$ $= -\frac{4m^3}{27} + n = 0$ $4m^3 = 27n$	2 marks for correct solution 1 mark for $x = \frac{2m}{3}$
c	$\int \frac{2x}{\sqrt{x^4 + 16}} dx$ let $u = x^2 \therefore du = 2xdx$ $= \int \frac{1}{\sqrt{u^2 + 4^2}} du$ $= \ln\left(u + \sqrt{u^2 + 16}\right) + C$ $= \ln\left(x^2 + \sqrt{x^4 + 16}\right) + C$	2 marks for correct solution 1 mark for $\int \frac{du}{\sqrt{u^2 + 1}}$

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d-i	$2x^2 - kx + 17 = 0$ $\alpha + \beta = \frac{k}{2}$ $\alpha = \frac{5}{2} + ib$ $\beta = \frac{5}{2} - ib$ $\therefore \alpha + \beta = 5$ $\therefore k = 10$ $x = \frac{k \pm \sqrt{k^2 - 136}}{4}$ $x = \frac{10 \pm \sqrt{100 - 136}}{4}$ $= \frac{5 \pm 3}{2} i$	2 marks for correct solution
d-ii	$\int_0^{\frac{\pi}{2}} \frac{dx}{2 + 2\sin x} = \frac{1}{2}$ $t = \tan \frac{x}{2} \therefore x = 2 \tan^{-1} t$ $dx = \frac{2}{1+t^2} dt \text{ also } \sin x = \frac{2t}{1+t^2}$ $x = 0 \quad t = 0$ $x = \frac{\pi}{2} \quad t = 1$ $\int_0^1 \frac{dt}{2 + 2 \times \frac{2t}{1+t^2}} \times \frac{2}{1+t^2}$ $= \int_0^1 \frac{dt}{t^2 + 2t + 1}$ $= \int_0^1 \frac{dt}{(t+1)^2}$ $= -1 \left[\frac{1}{t+1} \right]_0^1 = 1 - \frac{1}{2} = \frac{1}{2}$	2 marks for correct expression for transform. integral 1 mark for correct simplified expression for $\frac{1}{2 + 2\sin x}$ in terms of t

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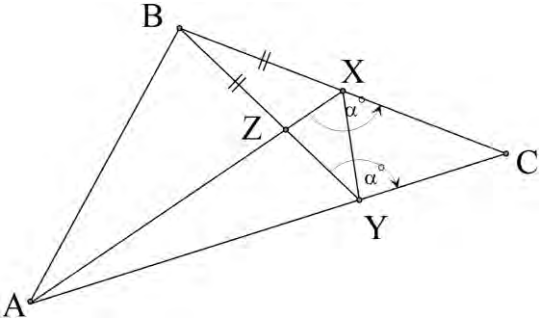
	Using $u = 2a - x \therefore -du = dx$ $x = 0 \quad u = 2a$ $x = a \quad u = a$ $- \int_{2a}^a f(u) du = \int_a^{2a} f(u) du$ nb $\int_a^{2a} f(u) du = \int_a^{2a} f(x) dx$ $\therefore \int_0^a f(x) dx + \int_a^{2a} f(x) dx = \int_0^{2a} f(x) dx$ $\therefore \int_0^{2a} f(x) dx = \int_0^a \{f(x) + f(2a - x)\} dx$ <i>Alternatively</i> $\begin{aligned} \int_0^a \{f(x) + f(2a - x)\} dx &= \left[F(x) - F(2a - x) \right]_0^a \\ &= [F(a) - F(a)] - [F(0) - F(2a)] \\ &= F(2a) - F(0) \end{aligned}$ $\int_0^{2a} f(x) dx = \left[F(x) \right]_0^{2a} = F(2a) - F(0)$	2 marks for correct solution 1 mark for decomposition of integral in terms of x and u.
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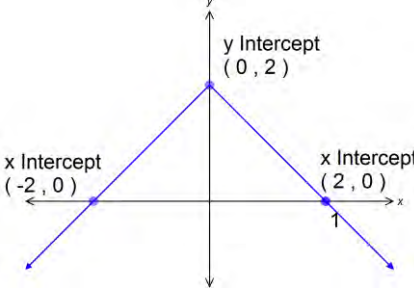
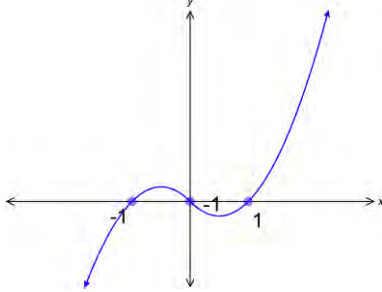
	$\int_0^{\pi} \frac{x}{2 + 2\sin x} dx = \int_0^{\frac{\pi}{2}} \frac{x}{2 + 2\sin x} + \frac{\pi - x}{2 + 2\sin(\pi - x)} dx$ $= \int_0^{\frac{\pi}{2}} \frac{x}{2 + 2\sin x} + \frac{\pi - x}{2 + 2\sin(x)} dx$ $= \int_0^{\frac{\pi}{2}} \frac{\pi}{2 + 2\sin x} dx$ $= \pi \int_0^{\frac{\pi}{2}} \frac{dx}{2 + 2\sin x}$ $= \pi \times \frac{1}{2} = \frac{\pi}{2}$	2 marks for correct solution 1 mark for correct transformation of integral
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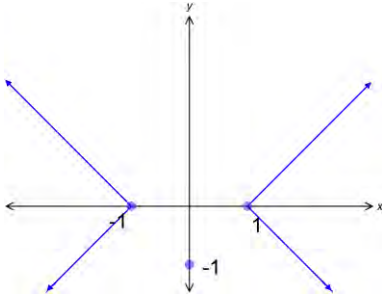
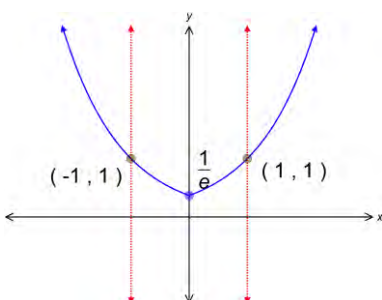
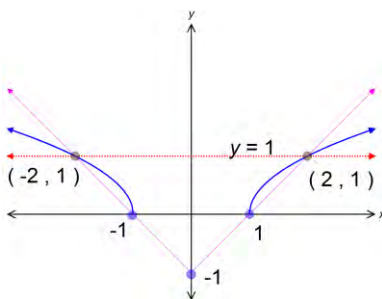
Question 13

a-i		1 mark – correct diagram with all given details
a-ii	$ABXY$ is a cyclic quadrilateral. Let $\angle AXC = \angle BYC = \alpha$ given $\angle BXZ = 180^\circ - \alpha$ Straight line $\angle AYB = 180^\circ - \alpha$ Straight line $\therefore \angle BXZ = \angle AYB$ $\therefore A, B, X, Y$ are concyclic (angles standing on same arc are equal) $\therefore ABXY$ is a cyclic quad.	2 marks – correct solution including all supporting statements. 1 mark – correct proof to two angles equal

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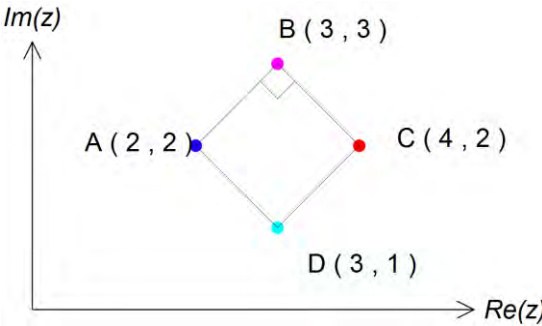
a-iii	prove AX bisects $\angle BAC$ $AX = XY$ given $\therefore \angle XBY = \angle XYB$ base $\angle$'s of isos. Δ $\angle XYB = \angle XAB$ angles standing on same arc equal $\angle XBY = \angle XAY$ angles standing on same arc equal $\therefore \angle XBY = \angle XAB$ $\therefore \angle XAY = \angle XAB$ $\therefore AX$ bisects $\angle BAY$	3 marks – fully explained correct solution 2 marks – correct logic but incomplete reasoning 1 mark – connection made between 3 equal angles
b-i	$y = 1 - f(x)$ 	1 mark – correct shape and points labelled
b-ii	$y = x.f(x)$ 	2 marks – correct shape and points labelled 1 mark – correct shape – no or incorrect corresponding points

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b-iii	$ y = f(x)$ 	2 marks – correct shape and points labelled 1 mark – correct shape – no or incorrect corresponding points
b-iv	$y = e^{f(x)}$ 	2 marks – correct shape and points labelled 1 mark – correct shape – no or incorrect corresponding points
b-v	$y = \sqrt{f(x)}$ 	2 marks – correct shape and points labelled and shape above and below y=1 1 mark – correct shape – no or incorrect corresponding points

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Q14

a-i	$x^3 - 11x + 20 = 0$ $x = 2 + i$ $\therefore (2 + i)^3 - 11(2 + i) + 20 = 0$ $8 + 12i + 6i^2 + i^3 - 22 - 11i + 20 = 0$ $(8 - 6 - 22 + 20) + (12i - i - 11i) = 0$ $0 = 0$ therefore $(z + i)$ is a root of the equation.	3 marks for correct solution 2 marks for $P(2 + i) = 0$ $\therefore P(2 - i) = 0$ 1 mark for $P(2 + i) = 0$
a-ii	Since all coefficients are real then the conjugate is also a root ie; $(z - i)$. therefore $x^2 - 2\text{Re}(x) + x ^2$ is a factor. $\therefore x^2 - 4x + 2^2 + 1^2 = x^2 - 4x + 5 \text{ is a factor}$ $x^3 - 11x + 20 = (x^2 - 4x + 5)(x + \alpha)$ $\therefore 20 = 5 \times \alpha$ $\alpha = 4$ $\therefore (x^3 - 11x + 20) = (x^2 - 4x + 5)(x + 4)$ $x = 2 + i \quad x = 2 - i \quad x = -4$	
b-i	Find the coordinates of B $(3 + 3i)$ and D $(3 + i)$ 	2 marks – correct solution 1 mark – correct B or C
b-ii	$A = 2 + 2i$ $iA = i(2 + 2i) = -2 + 2i$	1 mark – correct answer.

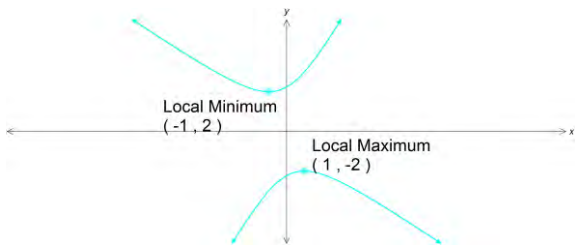
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c-i	$m = \frac{\frac{c}{p} - \frac{c}{q}}{cp - cq} = \frac{q-p}{pq} \times \frac{1}{p-q} = -\frac{1}{pq}$ $y - \frac{c}{p} = -\frac{1}{pq}(x - cp)$ $ypq - cq = -x + cp$ $x + ypq = (p + q)c$ Given the chord passes through A (6c , 4c)	2 marks for correct solution 1 mark for correct gradient
c-ii	Given the chord passes through A (6c , 4c) $x + ypq = (p + q)c$ $6c + 4cpq = (p + q)c$ $6 + 4pq = (p + q)$	1 mark – correct solution
c-iii	$x = \frac{c(p+q)}{2} \qquad y = \frac{c(p+q)}{2pq} = \frac{x}{pq}$ $\therefore \frac{x}{y} = pq$ also $4pq + 6 = p + q$ $y = \frac{c(4pq + 6)}{2pq}$ $ypq = \frac{c}{2}(4pq + 6)$ $y \times \frac{x}{y} = \frac{c}{2} \left(\frac{4x}{y} + 6 \right)$ $xy = 2cx + 3cy$ $(x - 3c)y = 2cx$ $y = \frac{2cx}{x - 3c}$	2 marks for correct solution 1 mark for correct x or y coordinate of midpoint.

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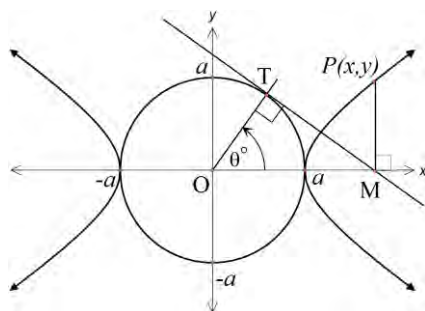
d-i	$x = \frac{c(p+q)}{2} \qquad y = \frac{c(p+q)}{2pq} = \frac{x}{pq}$ $\therefore \frac{x}{y} = pq$ also $4pq + 6 = p + q$ $x = \frac{c}{2}(4pq + 6)$ $x = \frac{2cx}{y} + 3c$ $x - 3c = \frac{2cx}{y}$ $y = \frac{2cx}{x - 3c}$ <i>CHECK</i> $ypq = \frac{c}{2}(4pq + 6)$ $y \times \frac{x}{y} = \frac{c}{2} \left(\frac{4x}{y} + 6 \right)$ $xy = 2cx + 3cy$ $(x - 3c)y = 2cx$ $y = \frac{2cx}{x - 3c}$	
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d-ii	$x^2 - y^2 + xy + 5 = 0$ $2x - \frac{2ydy}{dx} + y + \frac{xdy}{dx} = 0$ $(x - 2y)\frac{dy}{dx} = -(2x + y)$ $\frac{dy}{dx} = -\frac{2x + y}{x - 2y}$	2 marks for correct solution 1 mark for correct derivative expression
	Zero gradient at $-2x = y$ $x^2 - y^2 + xy + 5 = 0$ $x^2 - 4x^2 - 2x^2 + 5 = 0$ $-5x^2 + 5 = 0$ $x = \pm 1$ $\therefore -2x = y$ $x = 1 \quad y = -2$ $x = -1 \quad y = 2$ 	2 marks for correct solution 1 mark ofr correct equation from $\frac{dy}{dx} = 0$

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Question 15



$$OT = a$$

$$\sec\theta = \frac{OM}{a} \Rightarrow OM = a\sec\theta$$

$$H: \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$$\therefore \frac{a^2 \sec^2\theta}{a^2} - \frac{y^2}{b^2} = 1$$

$$\sec^2\theta - 1 = \frac{y^2}{b^2}$$

$$\tan^2\theta = \frac{y^2}{b^2}$$

$$\therefore y = b\tan\theta \text{ (first quadrant)}$$

2 marks – fully demonstrated.

1 mark – demonstrated by substitution into equation.

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Given that $\theta + \phi = \frac{\pi}{2}$ and $\theta \neq \frac{\pi}{2}$, show that the chord PQ has the equation $ay = b(\cos\theta + \sin\theta)x - ab$ $m = \frac{b(\tan\theta - \tan\phi)}{a(\sec\theta - \sec\phi)}$ $= \frac{b}{a} \left\{ \frac{\sin\theta}{\cos\theta} - \frac{\sin\phi}{\cos\phi} \right\} \div \left\{ \frac{1}{\cos\theta} - \frac{1}{\cos\phi} \right\}$ $= \frac{b}{a} \left\{ \frac{\sin\theta\cos\phi - \sin\phi\cos\theta}{\cos\theta\cos\phi} \right\} \times \left\{ \frac{\cos\theta\cos\phi}{\cos\phi - \cos\theta} \right\}$ $= \frac{b \sin(\theta - \phi)}{a\cos\phi - \cos\theta}$ $n\theta \phi = \frac{\pi}{2} - \theta \quad \therefore \quad \cos\phi = \sin\theta$ $= \frac{b \sin\left(\theta - \left(\frac{\pi}{2} - \theta\right)\right)}{a(\sin\theta - \cos\theta)} = \frac{b \sin\left(-\left(\frac{\pi}{2} - 2\theta\right)\right)}{a(\sin\theta - \cos\theta)}$ $= \frac{b(-\cos 2\theta)}{a(\sin\theta - \cos\theta)}$ $= \frac{-b(\cos^2\theta - \sin^2\theta)}{a(\sin\theta - \cos\theta)}$ $= \frac{-b(\cos\theta - \sin\theta)(\cos\theta + \sin\theta)}{a(\sin\theta - \cos\theta)}$ $= \frac{b(\cos\theta + \sin\theta)}{a}$ $y - b\tan\theta = \frac{b(\cos\theta + \sin\theta)}{a}(x - a\sec\theta)$ $ay - ab\tan\theta = b(\cos\theta + \sin\theta)x - ab - ab\tan\theta$ $ay = b(\cos\theta + \sin\theta)x - ab$	4 marks – correct solution fully detailed. 3 marks – gradient equal to $m = \frac{b(\cos\theta + \sin\theta)}{a}$ 2 marks – some further simplification of gradient using complementary angles 1 mark – recognition and some use of complementary angles - initial partially simplified expression for gradient in terms of θ and ϕ
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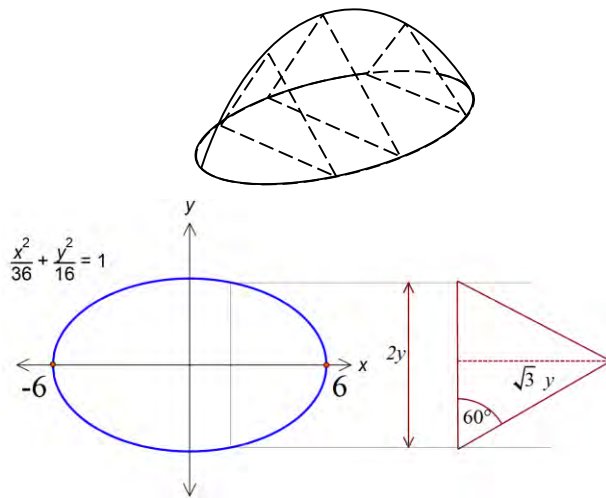
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	$ay = b(\cos\theta + \sin\theta)x - ab$ $x = 0 \quad y = -b$ Therefore for all chords PQ $x = 0$, y will equal $-b$ irrespective of angle θ ie. all chords will pass through the fixed point $(0, -b)$ since b is a constant.	2 marks – correct solution fully explained. 1 mark – partial explanation.
	$ay = b(\cos\theta + \sin\theta)x - ab$ asymptote $y = \pm \frac{b}{a}x$ as $\theta \rightarrow \frac{\pi}{2}$ $\cos\theta \rightarrow \cos\frac{\pi}{2} = 0$ $\sin\theta \rightarrow \sin\frac{\pi}{2} = 1$ $\therefore ay = b(0 + 1)x - ab$ $y = \frac{b}{a}x - b$ ie. parallel to asymptote	2 marks – correct solution – did require identification of equation to the asymptote $y = \pm \frac{b}{a}x$ 1 mark – correct equation without explanation.
	$x_1 + x_2 \geq 2\sqrt{x_1 x_2}$ $(x + y)^2 \geq 0 \quad \text{for all } x, y$ $x^2 + 2xy + y^2 \geq 0$ $x^2 + y^2 \geq 2xy$ Let $x \rightarrow \sqrt{x} \quad y \rightarrow \sqrt{y}$ $\therefore (\sqrt{x})^2 + (\sqrt{y})^2 \geq 2\sqrt{x}\sqrt{y}$ $x + y \geq 2\sqrt{xy}$	1 mark – a correct solution.

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	$\frac{x_1 + x_2 + x_3 + x_4}{4} \geq \sqrt[4]{x_1 x_2 x_3 x_4}$ $x_1 + x_2 \geq 2\sqrt{x_1 x_2}$ $x_3 + x_4 \geq 2\sqrt{x_3 x_4}$ $\therefore (x_1 + x_2) + (x_3 + x_4) \geq 2\sqrt{x_1 x_2} + 2\sqrt{x_3 x_4}$ $\frac{(x_1 + x_2) + (x_3 + x_4)}{2} \geq \sqrt{x_1 x_2} + \sqrt{x_3 x_4}$ $\sqrt{x_1 x_2} + \sqrt{x_3 x_4} \geq 2\sqrt{\sqrt{x_1 x_2} \sqrt{x_3 x_4}}$ $\therefore \frac{(x_1 + x_2) + (x_3 + x_4)}{2} \geq 2 \times \sqrt[4]{x_1 x_2 x_3 x_4}$ $\frac{(x_1 + x_2) + (x_3 + x_4)}{4} \geq \sqrt[4]{x_1 x_2 x_3 x_4}$	2 marks – correct solution 1 mark – correct to line (a)
	$\frac{x_1 + x_2 + x_3 + x_4}{4} > \sqrt[4]{x_1 x_2 x_3 x_4}$ $\text{let } \frac{(x_1 + x_2 + x_3)}{3} = x_4$ $\frac{x_1 + x_2 + x_3}{4} + \frac{x_1 + x_2 + x_3}{12} > \sqrt[4]{\frac{x_1 x_2 x_3 (x_1 + x_2 + x_3)}{3}}$ $\frac{4(x_1 + x_2 + x_3)}{12} \geq \sqrt[4]{\frac{x_1 x_2 x_3 (x_1 + x_2 + x_3)}{3}}$ $\frac{x_1 + x_2 + x_3}{3} \geq (x_1 x_2 x_3)^{\frac{1}{4}} \times \left(\frac{x_1 + x_2 + x_3}{3}\right)^{\frac{1}{4}} \quad (a)$ $\left(\frac{x_1 + x_2 + x_3}{3}\right)^{\frac{3}{4}} \geq (x_1 x_2 x_3)^{\frac{1}{4}}$ $\left(\frac{x_1 + x_2 + x_3}{3}\right)^3 \geq (x_1 x_2 x_3)$ $\left(\frac{x_1 + x_2 + x_3}{3}\right) \geq \sqrt[3]{(x_1 x_2 x_3)}$	2 marks – correct solution. 1 mark – correct to line (a)

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$$\text{Area of Triangle} = \frac{1}{2}bh = \frac{1}{2}2y\sqrt{3}y = \sqrt{3}y^2$$

$$\delta V = \sqrt{3}y^2 \delta x$$

$$V \approx 2 \times \lim_{\delta x \rightarrow 0} \sum_{x=0}^6 \sqrt{3}y^2 \delta x$$

$$= 2\sqrt{3} \int_0^6 y^2 dx$$

$$= 2\sqrt{3} \int_0^6 16 \left(1 - \frac{x^2}{36}\right) dx$$

$$= 32\sqrt{3} \times \int_0^6 1 - \frac{x^2}{36} dx$$

$$= 32\sqrt{3} \left[x - \frac{x^3}{36} \right]_0^6$$

$$= 32\sqrt{3} \left(6 - \frac{6^3}{3 \times 36} \right)$$

$$= 32\sqrt{3} \times 4 = 128\sqrt{3} \text{ units}^3$$

4 marks for correct solution

3 marks

- **Correct primitive function**
- **Correct evaluation from incorrect δx**

2 marks for correct volume element and correct integral expression

1 mark for correct area expression

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$I_n = \int_0^1 x(1-x^5)^n dx$ $u = (1-x^5)^n \quad v' = x$ $u' = -5nx^4(1-x^5)^{n-1} \quad v = \frac{x^2}{2}$ $I_n = \left[\frac{x^2}{2} \times (1-x^5)^n \right]_0^1 - \int_0^1 -\frac{5n}{2} x^4(1-x^5)^{n-1} x^2 dx$ $= 0 + \frac{5n}{2} \int_0^1 x^6(1-x^5)^{n-1} dx$ $= \frac{5n}{2} \int_0^1 (-x)(-x^5)(1-x^5)^{n-1} dx$ $= \frac{5n}{2} \int_0^1 (-x)\{(1-x^5) - 1\}(1-x^5)^{n-1} dx$ $= \frac{5n}{2} \int_0^1 (-x)(1-x^5)^n - (-x)(1-x^5)^{n-1} dx$ $= -\frac{5n}{2} \int_0^1 (x)(1-x^5)^n - (x)(1-x^5)^{n-1} dx$ $I_n = -\frac{5n}{2}(I_n - I_{n-1})$ $I_n = -\frac{5n}{2}I_n + \frac{5n}{2}I_{n-1}$ $I_n + \frac{5n}{2}I_n = \frac{5n}{2}I_{n-1}$ $2I_n + 5nI_n = 5nI_{n-1}$ $(2+5n)I_n = 5nI_{n-1}$ $I_n = \frac{5n}{5n+2}I_{n-1}$	3 marks for correct solution 2 marks for correct decomposition of I_n using $x^6 = x\{(1-x)^5 - 1\}$ 1 mark for correct application fo IBP to obtain $\frac{5n}{2} \int_0^1 (1-5x)^{n-1} dx$
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b-ii	$\int_0^1 (-x)(1-x^5)^n - (-x)(1-x^5)^{n-1} dx$ $I_n = \frac{5n}{5n+2} I_{n-1}$ $I_{n-1} = \frac{5(n-1)}{5(n-1)+2} I_{n-2} = \frac{5n-5}{5n-3} I_{n-2}$ $I_{n-2} = \frac{5(n-2)}{5(n-2)+2} I_{n-3} = \frac{5n-10}{5n-8} I_{n-4}$ $\therefore I_{n-4} = \frac{5n-15}{5n-13} I_{n-4} \dots$ $I_2 = \frac{5 \times 2}{5 \times 2 + 2} I_1$ $I_1 = \frac{5}{5+2} I_0$ $I_0 = \int_0^1 x(1-x)^0 dx = \left[\frac{x^2}{2} \right]_0^1 = \frac{1}{2}$ $\therefore$ $I_n = \frac{5n}{5n+2} \times \frac{5n+5}{5n-3} \times \frac{5n-10}{5n-8} \times \dots \times \frac{10}{12} \times \frac{5}{7} \times \frac{1}{2}$ $= \frac{5(n) \times 5(n-1) \times 5(n-2) \times \dots \times 5(2) \times 5(1) \times 1}{2 \times 7 \times 12 \times \dots \times (5n+2)}$ $= \frac{5^n n!}{2 \times 7 \times 12 \times \dots (5n+2)}$	2 marks for correct solution 1 mark for correct expression of initial three terms Or 1 mark for evaluation of I_0
	$I_4 = \frac{5^4 \times 4!}{2 \times 7 \times 12 \times 17} = \frac{625}{119}$	1 mark for correct answer

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c	$\int \{f''(x) g(x) - f(x) g''(x)\} dx = f'(x) g(x) - f(x) g'(x)$ $\therefore \int \{f''(x) g(x) - f(x) g''(x)\} dx = \int f''(x) g(x) dx - \int f(x) g''(x) dx$ $\int f''(x) g(x) dx$ $u = g(x) \qquad v' = f''(x)$ $u' = g'(x) \qquad v = f'(x)$ $\therefore \int f''(x) g(x) dx = g(x) f'(x) - \int f'(x) g'(x) dx \quad \textcircled{1}$ $\int f(x) g''(x) dx$ $u = f(x) \qquad v' = g''(x)$ $u' = f'(x) \qquad v = g'(x)$ $\therefore \int f(x) g''(x) dx = g'(x) f(x) - \int f'(x) g'(x) dx \quad \textcircled{2}$ $\textcircled{1} - \textcircled{2}$ $g(x) f'(x) - \int f'(x) g'(x) dx - \left\{ g'(x) f(x) - \int f'(x) g'(x) dx \right\}$ $= g(x) f'(x) - g'(x) f(x)$ $\therefore \int \{f''(x) g(x) - f(x) g''(x)\} dx = f'(x) g(x) - f(x) g'(x)$	2 marks for correct solution 1 mark for - Correct application of product rule to RHS - Correct application of IBP to LHS
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	$f(x) = \sin 3x \qquad g(x) = \cos 2x$ $f'(x) = 3 \cos 3x \quad g'(x) = -2 \sin 2x$ $f''(x) = -9 \sin 3x \qquad g''(x) = -4 \cos 2x$ $\int f'(x) g(x) - f(x) g''(x) dx$ $= \int -9 \sin 3x \cos 2x - \sin^3 x \times (-4 \cos 2x) dx$ $= \int -5 \sin 3x \cos 2x dx$ $= -5 \int \sin 3x \cos 2x dx$ From the previous part (i) therefore	3 marks for correct solution
c-iii	$\int f'(x) g(x) - f(x) g''(x) dx = f'(x) g(x) - f(x) g'(x)$ $= -5 \int \sin 3x \cos 2x dx$ $= -5 \{ 3 \cos 3x \cos 2x - \sin 3x \times (-2 \sin 2x) \}$ $= -5 \left[3 \cos 3x \cos 2x + 2 \sin 3x \sin 2x \right]_0^{\frac{\pi}{4}}$ $= -5 \left\{ \left(3 \times -\frac{1}{\sqrt{2}} \times 0 + 2 \times \frac{1}{\sqrt{2}} \times 1 \right) - (3 \times 1 \times 1 + 2 \times 0 \times 0) \right\}$ $= -5 \left(\frac{2}{\sqrt{2}} - 3 \right)$ $\therefore \int \sin 3x \cos 2x = \frac{1}{5} \times \left(3 - \frac{2}{\sqrt{2}} \right)$	2 marks for substantial progress towards solution 1 mark for relevant initial progress to a solution.

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Or Method 2

$$\int_0^{\frac{\pi}{4}} \sin 3x \cos 2x \, dx$$

$$\int \{f''(x) g(x) - f(x) g''(x)\} \, dx = f'(x) g(x) - f(x) g'(x)$$

$$u = \sin 3x \quad v' = \cos 2x$$

$$u' = 3\cos 3x \quad v = \frac{1}{2}\sin 2x$$

$$I = \frac{1}{2} \sin 3x \sin 2x - \frac{3}{2} \int \cos 3x \sin 2x$$

$$u = \cos 3x \quad v' = \sin 2x$$

$$u' = -3\sin 3x \quad v = -\frac{1}{2} \cos 2x$$

$$I_2 = -\frac{1}{2} \cos 3x \cos 2x - \frac{3}{2} \int \sin 3x \cos 2x \, dx$$

$$= -\frac{1}{2} \cos 3x \cos 2x - \frac{3}{2} I$$

$$I = \frac{1}{2} \sin 3x \sin 2x - \frac{3}{2} \left\{ -\frac{1}{2} \cos 3x \cos 2x - \frac{3}{2} I \right\}$$

$$= \frac{1}{2} \sin 3x \sin 2x + \frac{3}{4} \cos 3x \cos 2x + \frac{9}{4} I$$

$$-\frac{5}{4} I = \left[\frac{1}{2} \sin 3x \sin 2x + \frac{3}{4} \cos 3x \cos 2x \right]_0^{\frac{\pi}{4}}$$

$$-5 I = \left[2 \sin 3x \sin 2x + 3 \cos 3x \cos 2x \right]_0^{\frac{\pi}{4}}$$

$$-5 I = \left(2 \times \frac{1}{\sqrt{2}} \times 1 + 3 \times -\frac{1}{\sqrt{2}} \times 0 \right) - (2 \times 0 \times 0 + 3 \times 1 \times 1)$$

$$I = \frac{-1}{5} \left(\frac{2}{\sqrt{2}} - 3 \right) = \frac{1}{5} \left(3 - \frac{2}{\sqrt{2}} \right)$$